

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Title – AgriLink AI

Domain of Project – Economic Development

Student Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
SHRUTI PATEL		shrutime0110@gmail.com	6357977129

Problem Statement - Agricultural Market Access

The Challenge

Farmers often struggle with **limited market information, credible buyer access, and informed selling decisions**. Without **timely price insights and simple AI-powered guidance**, they may find it difficult to decide **when, where, and at what value to sell their produce**.

The Objective

Build an **AI-powered AgriLink AI platform** that delivers **market intelligence, buyer connections, and selling guidance** for farmers. The solution should:

- **Price Intelligence** – Provide mandi prices, market trends, and price forecasts.
- **Buyer Discovery** – Connect farmers with suitable and credible buyers.
- **Selling & Storage Advisory** – Support informed decisions on whether to sell or store produce.
- **Quality Guidance** – Help farmers understand quality factors and buyer requirements.

Multi-Agent System :–

- **Market Intelligence Agent** – Provides price and trend insights.
- **Buyer Matching Agent** – Connects farmers with relevant buyers.
- **Selling Advisory Agent** – Supports sell-or-store decisions.
- **Quality & Income Agent** – Provides quality guidance and income visibility.
- **Farmer Dashboard** – Show prices, buyers, recommendations, and income insights in one place.

Proposed Solution - AgriLink AI Platform

The proposed solution is an **AI-assisted market intelligence and direct-trade platform** designed for **cotton and groundnut farmers**. AgriLink AI brings together market prices, forecasting, buyer discovery, selling guidance, quality assessment, and conversational AI into a single intelligent platform.

At its core, the solution provides **mandi-wise price intelligence** by presenting current prices, historical trends, and relevant market insights. AI-powered analysis helps farmers understand market movements and make more informed decisions.

The system consists of six key components:

- 1) **Price Intelligence** provides mandi-wise current prices, historical trends, filtering, and AI-generated market explanations.
- 2) **Forecasting Agent** delivers an indicative **7–14 day price outlook** based on market trends, confidence signals, and available market factors.
- 3) **Buyer Matching** connects farmers with relevant buyers by comparing buyer types, offers, requirements, and response or fit signals.
- 4) **Selling Advisor** helps farmers evaluate different options, such as **selling immediately, storing produce, or exploring direct-buyer opportunities**, based on their individual needs.
- 5) **Quality + Income** provides quality assessment guidance and estimated sale or income breakdowns to support better decision-making.
- 6) **Granite AI Assistant** offers conversational guidance in **English, Gujarati, and Hindi**, making the platform more accessible and easier for farmers to use.

The solution uses **IBM Granite as the AI reasoning layer**, while **LangFlow supports the orchestration of the agentic workflow**. Together, these components create a modular system where different AI capabilities work together to provide actionable market guidance.

Overall, KisanSetu delivers an **end-to-end intelligent agricultural market platform** that helps farmers access market information, connect with buyers, evaluate selling strategies, and make more confident decisions.

Technology Used

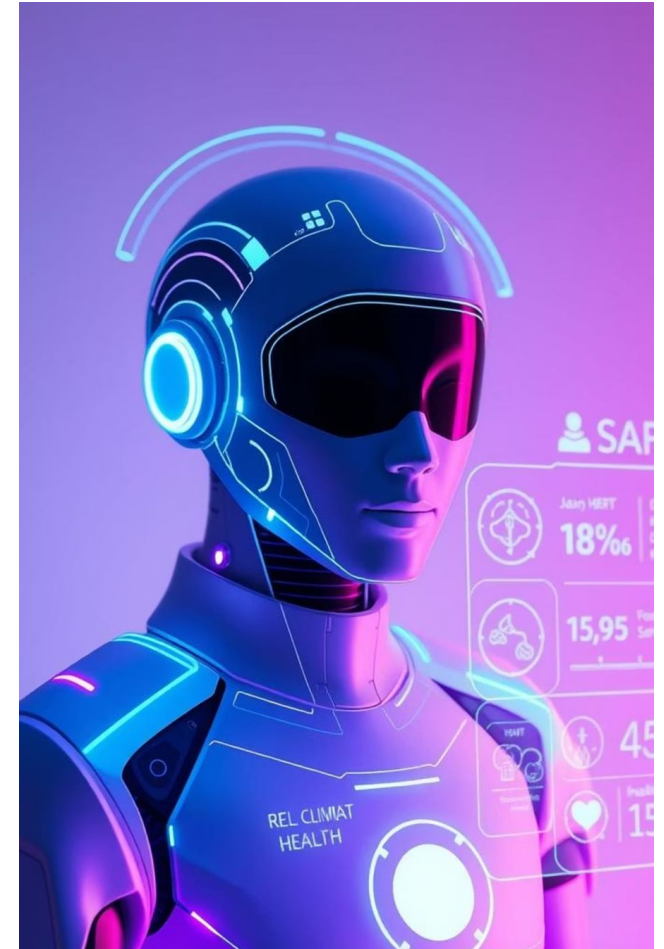
- **LangFlow Platform** – For designing and orchestrating multi-agent AI workflows visually.
- **IBM Granite Model** – For natural language understanding, reasoning, and intelligent decision support.
- **IBM Cloud / watsonx.ai** – For building, deploying, and scaling the AI solution with model deployment and AI governance.
- **RAG (Retrieval-Augmented Generation)** – To retrieve relevant market and agricultural data first, then generate informed AI responses.
- **Vector Database** – For storing embeddings and enabling fast retrieval of relevant market information.

Langflow component Used -

- 1) **Chat Input** – Allows farmers to enter queries related to **crop prices, buyers, storage, selling decisions, and market insights**.
- 2) **Context / Retrieval** – Retrieves relevant **market, crop, and farmer-specific information** to provide context for AI responses.
- 3) **IBM Granite LLM / Agent** – Processes the user's query and reasons over the available context to generate **actionable market insights and recommendations**.
- 4) **Chat Output** – Displays AI-generated responses such as **price insights, forecasts, buyer suggestions, and selling recommendations**.
- 5) **Specialized Tools / Modules** – Connects the AI assistant to modules such as **Price Intelligence, Forecasting, Buyer Matching, Selling Advisor, Quality, and Income**.
- 6) **Conversation Context / Memory** – Maintains the user's **language, query context, and interaction flow** for a more consistent conversational experience.

Role of Agentic AI in the solution

- **Agentic AI** enables AgriLink AI to function as an intelligent agricultural market decision-support system rather than a simple chatbot. Using **IBM Granite models**, it understands farmer queries and uses relevant market and crop information to provide actionable guidance.
- The system supports specialized capabilities such as **mandi price intelligence, price forecasting, direct buyer–farmer matching, storage and selling advice, quality grading, and income analysis**. Each capability addresses a specific part of the farmer's decision-making process.
- Agentic AI brings **context awareness** by considering factors such as the farmer's crop, location, quantity, quality, and current market conditions. It can then help compare different options, such as **selling now, storing produce, or choosing a suitable buyer**.
- It also supports **intelligent reasoning and recommendations**, converting available market information into simple and actionable insights for farmers.
- Overall, Agentic AI makes AgriLink AI **context-aware, goal-driven, and decision-focused**, helping cotton and groundnut farmers make more informed market and selling decisions.



Novelty and Uniqueness

The proposed **AgriLink AI** platform stands out by combining **Agentic AI, market intelligence, and decision support** into a single solution for cotton and groundnut farmers.

- **Farmer-First Decision Intelligence** – Combines **price, timing, buyer, and quality insights** to support complete selling decisions.
- **Direct Trade Discovery** – Goes beyond market information by helping farmers identify and compare **relevant buyer opportunities**.
- **Scenario-Based Advice** – Supports decisions such as **sell now, store, or explore direct buyers** by considering price, quantity, and practical constraints.
- **Multilingual Accessibility** – Provides a simpler experience through support for **English, Gujarati, and Hindi** interactions.
- **Explainable AI Experience** – Recommendations include **reasons, trade-offs, and verification guidance**, rather than acting as opaque suggestions.
- **End-to-End Income View** – Connects **market intelligence, quality assessment, and income insights** to support the farmer throughout the selling journey.

This makes AgriLink AI a **decision-focused agricultural intelligence platform**, not just a price information or buyer discovery tool.

GitHub Repository Link

Public GitHub Repository:

<https://github.com/ShrrutiPatel/AgrilinkAI->

Repository: AgriLink AI

The repository contains the following project files:

- **app.json** – Configuration and project application details for **AgriLink AI**.
- **problemstatement.pdf** – Problem statement for the **AI-Powered Cotton & Groundnut Market Linkage Platform**.
- **projectpresentation.pptx** – Complete presentation of the **AgriLink AI solution**.
- **Source Code & Supporting Files** – Files related to market intelligence, price forecasting, buyer–farmer matching, selling/storage advisory, quality guidance, dashboard, and other supporting components.

Future Scope

1. **Live Market Data Integration –**
Integrate live mandi and market data for **real-time prices and updated insights**.
2. **Voice & Regional Language Support –**
Enable **voice interaction and more regional languages** for better farmer accessibility.
3. **Vision-Based Quality Assessment –**
Use image-based AI to assist with **crop quality and grading analysis**.
4. **Smart Alerts & Notifications –**
Notify farmers about **price changes, buyer responses, and market updates**.
5. **Trust & Verification Layer –**
Add **buyer verification and transaction records** for safer direct trade.

Certificates Earned:

IBM SkillsBuild

Completion Certificate



This certificate is presented to

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Getting Started with Artificial Intelligence

(PLAN-E624C2604060)

According to the Your Learning Builder - Plans system of record

Completion date: 31 Aug 2026 (GMT)

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IBM **SkillsBuild**



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Certificates Earned:

IBM SkillsBuild

Completion Certificate



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for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 01 Sep 2026 (GMT)

Learning hours: 30 mins

Thank You!

Thank you for your time and interest.